client authentication method supported by your AS. The parameters are URL-encoded as for other OAuth2 requests. The AS returns a 200 OK response if the request is successful, with the device code, user code, and verification URI in JSON format. Navigate to src/main/java/com/manning/apisecurityinaction and create a new file named DeviceGrantClient.java. Create a new public class in the file with the same name and add the method from listing 13.16 to the file. You'll need the following imports at the top of the file:

```
import org.json.JSONObject;
import java.net.*;
import java.net.http.*;
import java.net.http.HttpRequest.BodyPublishers;
import java.net.http.HttpResponse.BodyHandlers;
import java.util.concurrent.TimeUnit;
import static java.nio.charset.StandardCharsets.UTF_8;
```

Listing 13.16 Starting a device authorization grant flow

```
private static final HttpClient httpClient =
HttpClient.newHttpClient();

private static JSONObject beginDeviceAuthorization(
        String clientId, String scope) throws Exception {
    var form = "client_id=" + URLEncoder.encode(clientId,
UTF_8) +    #A
            "&scope=" + URLEncoder.encode(scope, UTF_8);
#A
    var request = HttpRequest.newBuilder()    #A
            .header("Content-Type",     #A
```